

NAME : SHAH FAISAL

ROLL NO : 23I-0058

SECTION : BSE – 6A

Computer Networks Lab 06

BSSE SPRING 2026

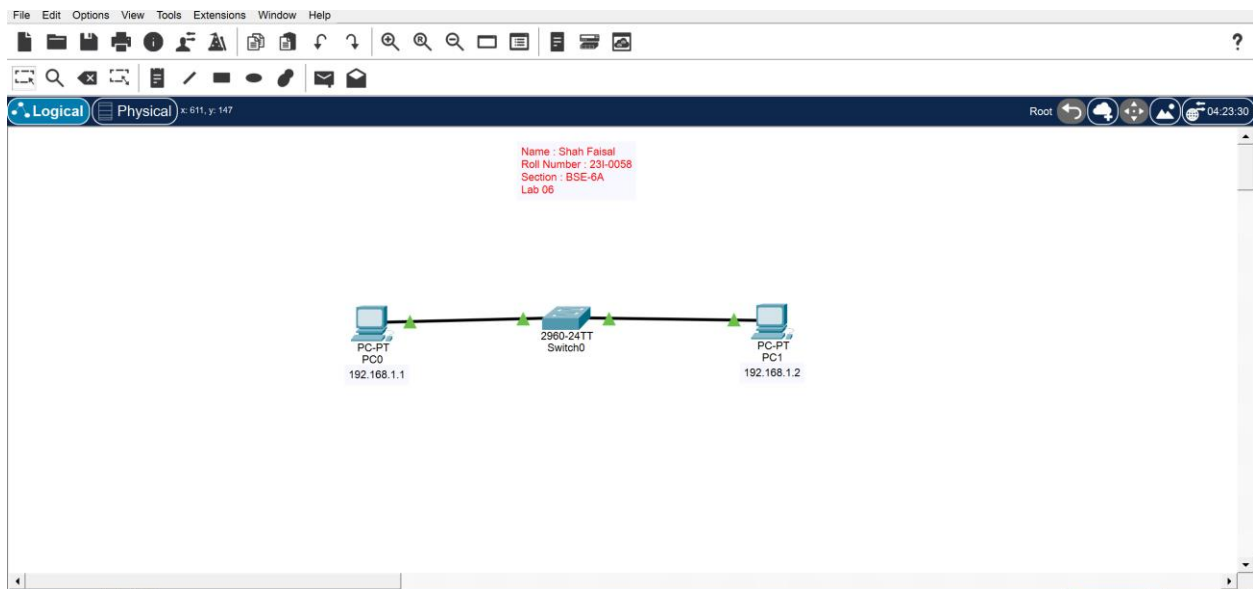
LAB 06

Task 1: Stop-and-Wait ARQ Simulation (5 Marks)

Simulate the Stop-and-Wait ARQ protocol by setting up two PCs connected to a switch. PC0 will have the IP address 192.168.1.1, and PC1 will have 192.168.1.2. In Cisco Packet Tracer, place the two PCs and a switch, then connect them using straight-through cables. Assign the specified IP addresses to each PC. Open the Command Prompt on PC0 and send a ping request to PC1. Switch to Simulation Mode to observe the packet flow.

To introduce an error, manually turn off the connection between PC1 (your screenshot should clearly show which interface you are doing this for) and the switch after the request is sent and turn it back on after a few seconds. Observe how PC0 handles the missing acknowledgment (ACK) and retransmits the data.

Question: What happens if an acknowledgment (ACK) is lost during Stop-and-Wait ARQ?
Explain how the sender handles this situation.



PC0

Physical Config **Desktop** Programming Attributes

Command Prompt

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

***Untitled - Notepad**
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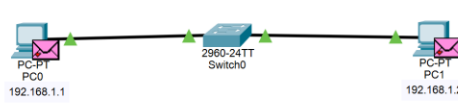
Top

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical **Physical** x: 796, y: 315

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Simulation Panel
 Event List

Vis.	Time(sec)	Last Device
	0.995	--
	0.996	Switch0
	0.996	Switch0
	2.993	--
	2.994	Switch0
	2.994	Switch0

 Reset Simulation ☒ Constant Delay Captured to: 2.994 s
 Play Controls
 Event List Filters - Visible Events
 ACL Filter, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
 Edit Filters Show All/None
 Event List Realtime Simulation

Time: 00:10:48.744 PLAY CONTROLS

Switch0

Physical
Config
CLI
Attributes

GLOBAL
 Settings
 Algorithm Settings
SWITCHING
 VLAN Database
INTERFACE
 FastEthernet0/1
 FastEthernet0/2
 FastEthernet0/3
 FastEthernet0/4
 FastEthernet0/5
 FastEthernet0/6
 FastEthernet0/7
 FastEthernet0/8

FastEthernet0/2
 Port Status ☐ On
 Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
 Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto
 Access VLAN
 Tx Ring Limit

Equivalent IOS Commands

```

Switch(config-if)#exit
Switch(config)#interface FastEthernet0/2
Switch(config-if)#shutdown
Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
no shutdown
Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
          
```

*Untitled - Notepad
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After disabling port status

Logical
Physical
x: 613, y: 381

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PC-PT
PC0
192.168.1.1

2960-24TT
Switch0

PC-PT
PC1
192.168.1.2

Simulation Panel

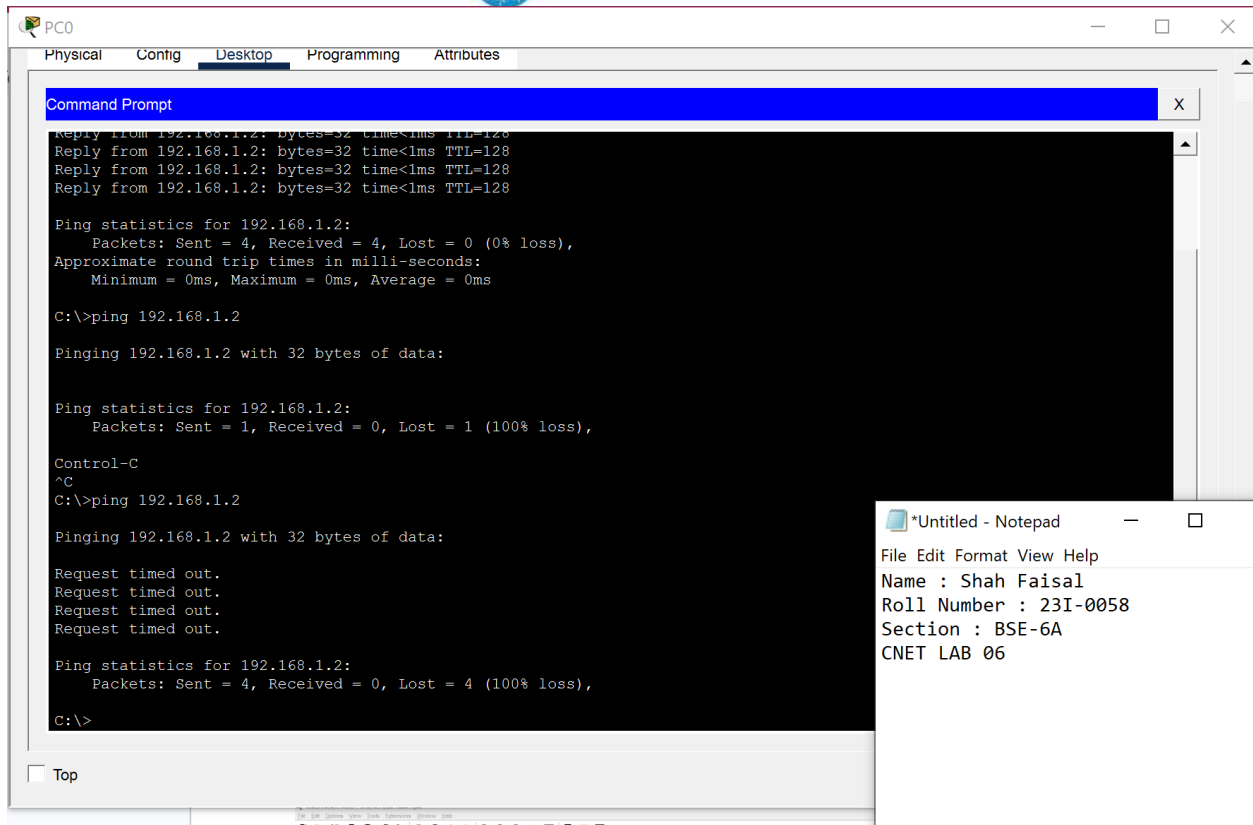
Vis.	Time(sec)	Last Device
	16.699	--
	16.700	Switch0
	18.698	--
	18.699	Switch0
	20.696	--
	20.697	Switch0
	20.707	--

Reset Simulation ☒ Constant Delay Captured to: 20.707 s

Play Controls

Event List Filters - Visible Events
 ACL Filter, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None



What happens if an acknowledgment (ACK) is lost during Stop-and-Wait ARQ? Explain how the sender handles this situation.

If an ACK is lost in Stop-and-Wait ARQ, the sender does not receive confirmation before the timer expires, so it assumes the frame was lost.

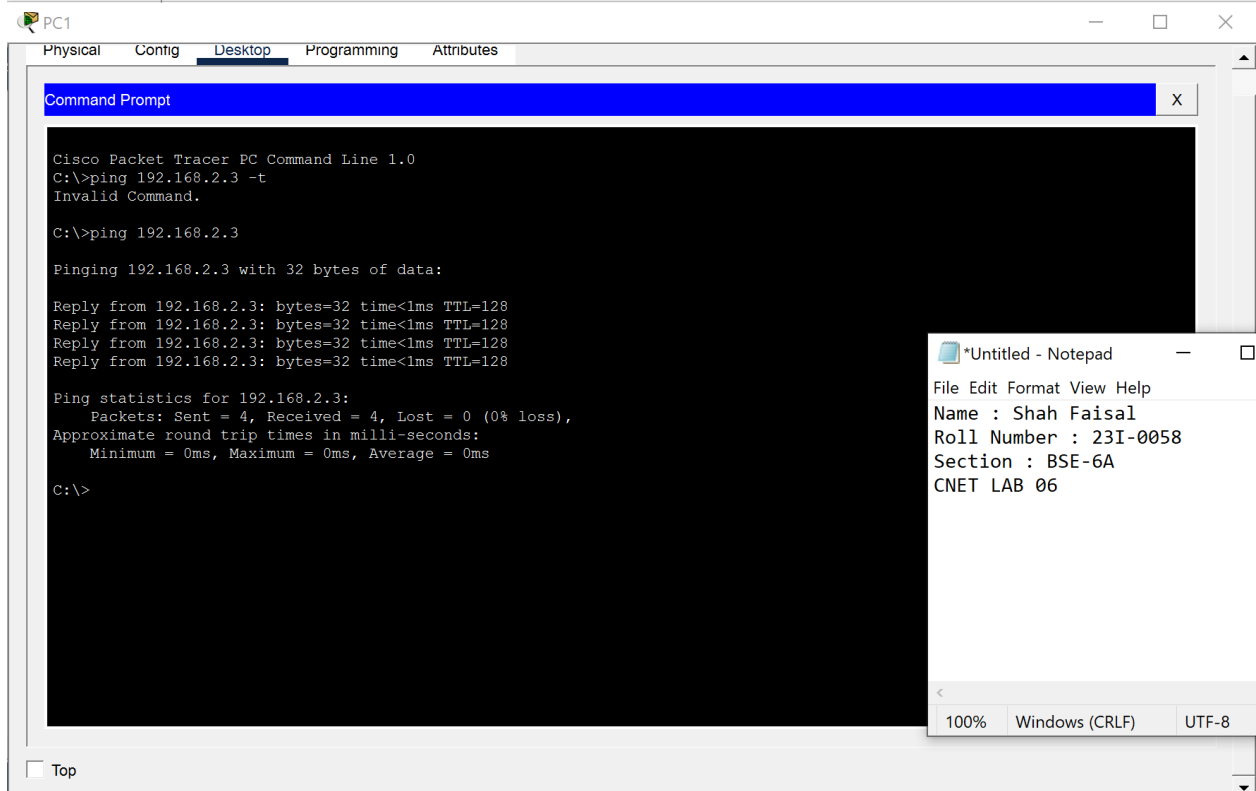
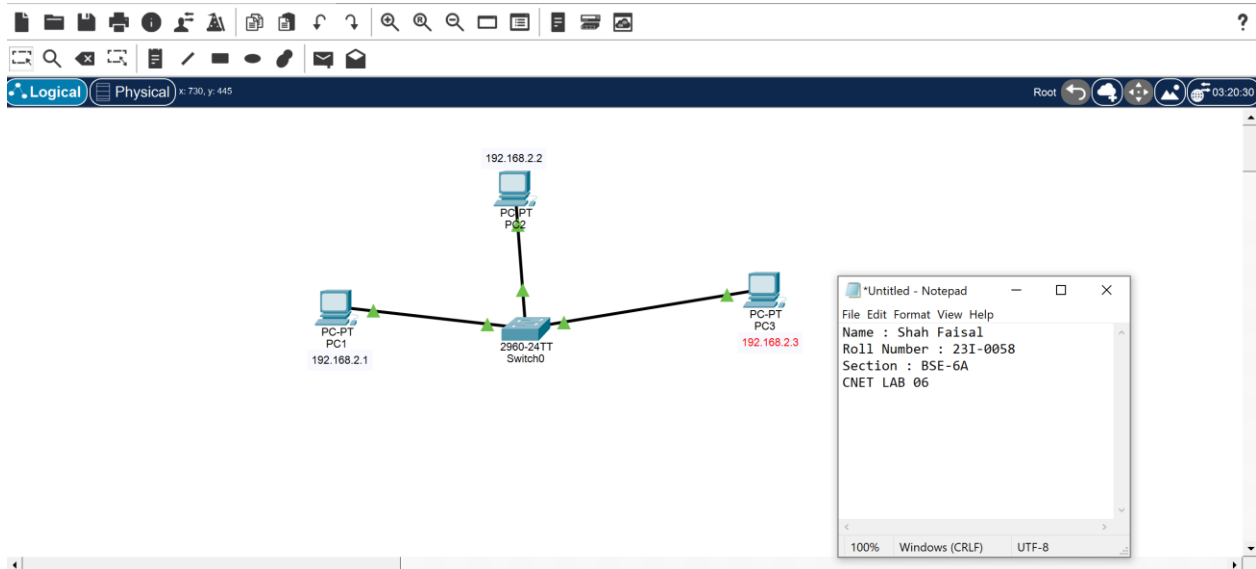
After the timeout, the sender retransmits the same frame with the same sequence number. The receiver detects the duplicate frame, discards it, and sends the ACK again, keeping communication reliable.

Task 2: Go-Back-N ARQ Simulation (5 Marks)

Simulate the Go-Back-N ARQ protocol by setting up three PCs connected through a switch. PC0 will have the IP address 192.168.2.1, PC1 will use 192.168.2.2, and PC2 will have 192.168.2.3. In Cisco Packet Tracer, place three PCs and a switch, then connect all PCs to the switch using straight-through cables. Assign the specified IP addresses to each PC. Open the Command Prompt on PC0 and send continuous ping requests to PC2 while monitoring the transmission in Simulation Mode. To introduce an error, disable PC2's connection to the switch (your screenshot

should clearly show which interface you are doing this for) after the first two packets have been sent and re-enable it after a few seconds. Observe how multiple frames are retransmitted once the connection is restored, as Go-Back-N requires retransmission of all frames after a lost or corrupted one.

Question: What is the impact of using Go-Back-N ARQ on network efficiency when multiple frames are lost in a row?



The screenshot shows the PC Command Line window for PC1. The output of the command `C:\>ping 192.168.2.3` is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3 -t
Invalid Command.

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

A Notepad window titled "Untitled - Notepad" is also open on the right, displaying the same student information as in the previous screenshot.

Logical

Physical

x: 634, y: 231

Root

05:06:00

192.168.2.2

PC-PT PC2

192.168.2.1

PC-PT PC1

2960-24TT

Switch0

192.168.2.3

PC-PT PC3

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Simulation Panel

Event List

Vis.	Time(sec)	Last Device
6.543	--	
6.544		Switch0
7.993	--	
7.994		Switch0
7.994		Switch0
7.994		Switch0

Reset Simulation
 ☒ Constant Delay
 Captured to: 7.994 s

Play Controls

Event List Filters - Visible Events
 ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Switch0

Physical

Config

CLI

Attributes

GLOBAL

Settings
 Algorithm Settings

 SWITCHING

VLAN Database

 INTERFACE

FastEthernet0/1
 FastEthernet0/2
 FastEthernet0/3
 FastEthernet0/4
 FastEthernet0/5
 FastEthernet0/6
 FastEthernet0/7
 FastEthernet0/8

FastEthernet0/3

Port Status
 Link Speed
 Duplex
 Access
 Tx Ring Limit

VLAN
 1
 10

☐ On
☒ 100 Mbps ☐ 10 Mbps ☒ Auto
☐ Half Duplex ☒ Full Duplex ☒ Auto

Equivalent IOS Commands


```

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface FastEthernet0/3
      
```

*Untitled - Notepad

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PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.2.3 -t
Invalid Command.

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Ping statistics for 192.168.2.3:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),

Control-C
^C
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:
```

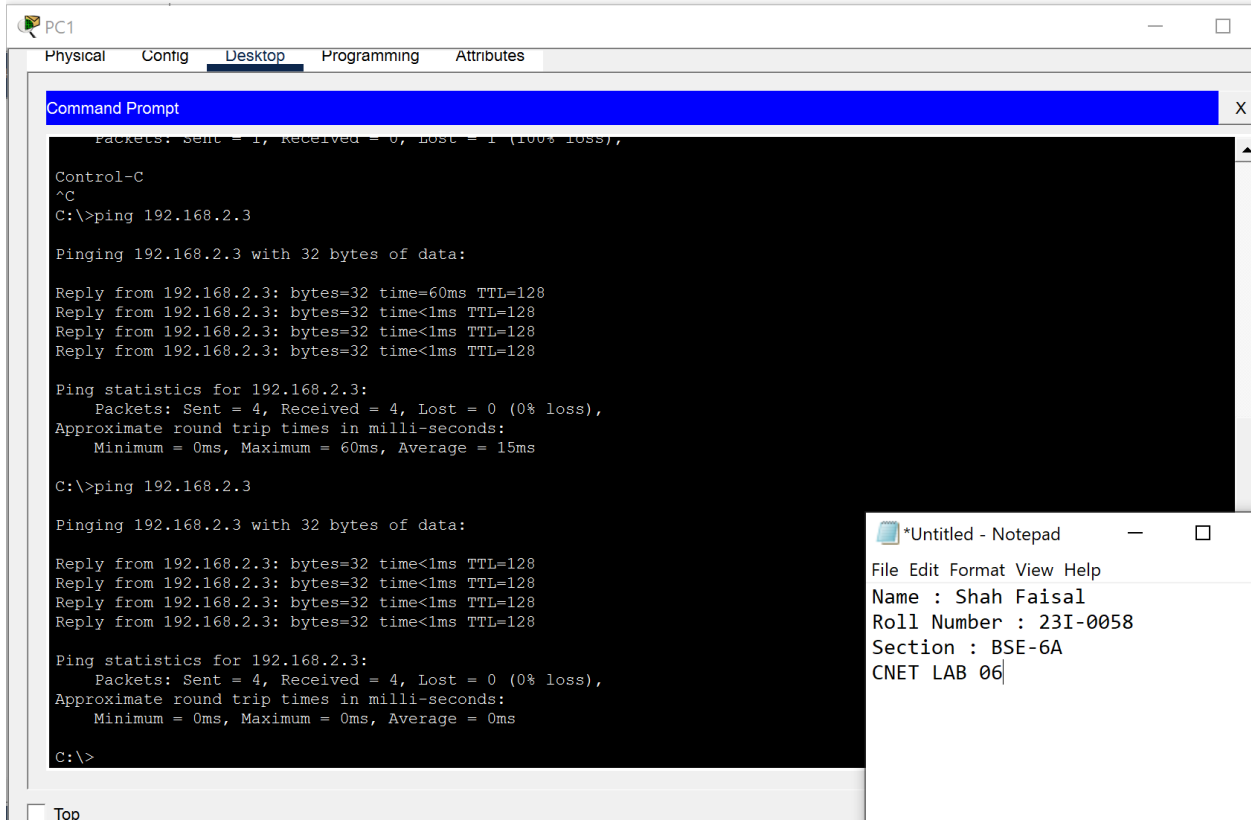
Top

*Untitled - Notepad

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AFTER TURNING IT ON



The screenshot shows a PC1 desktop environment. The 'Desktop' tab is selected in the top menu. A 'Command Prompt' window is open, displaying the following text:

```

Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),

Control-C
^C
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=60ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 60ms, Average = 15ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
  
```

A 'Notepad' window titled '*Untitled - Notepad' is also open, containing the following text:

```

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```

Question: What is the impact of using Go-Back-N ARQ on network efficiency when multiple frames are lost in a row?

When multiple frames are lost in Go-Back-N ARQ, the sender must retransmit the lost frame and all frames sent after it, even if some were received correctly, causing bandwidth waste. This leads to lower efficiency and throughput, because repeated retransmissions increase delay and reduce the effective data transfer rate.

As losses increase, the window keeps resetting, which adds more latency and makes the network much less efficient.

Task 3: Selective Repeat Protocol Simulation (5 Marks)

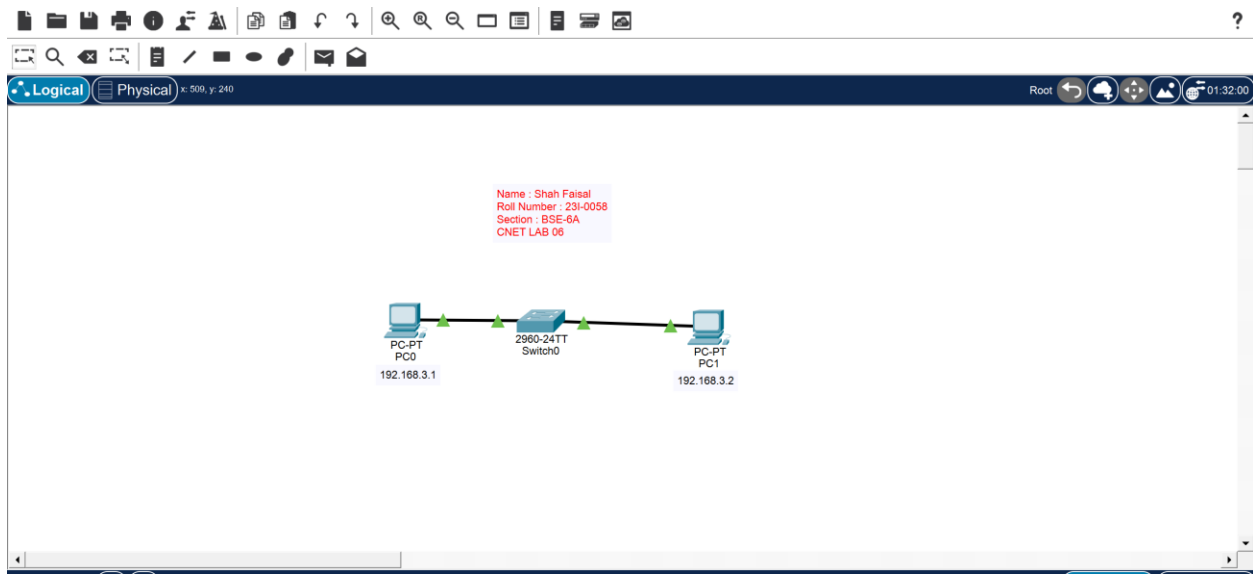
Simulate the Sliding Window Protocol using a switch and two PCs. PC0 will have the IP address 192.168.3.1, and PC1 will have 192.168.3.2. In Cisco Packet Tracer, set up a switch and connect both PCs to it using straight-through cables. Assign the specified IP addresses to the PCs and configure continuous ping requests from PC0 to PC1 while running the simulation in Simulation Mode. The sliding window size should be considered as 4, meaning you have to send 4 packets simultaneously.

To introduce an error, disable PC1's Ethernet port right after receiving two packets and re-enable it after a few seconds (your screenshot should clearly show which interface you are doing this for). Observe how only the unacknowledged packets are retransmitted,

demonstrating how the Sliding Window Protocol optimizes network efficiency by avoiding unnecessary retransmissions.

Question 1: What potential issues could arise if the sender's window size is too large compared to the network capacity, and how might this affect overall performance?

Question 2: How does the fixed sliding window size of 4 in this simulation limit the accuracy of real-world protocol behavior, and what changes could be made to better reflect real network conditions?



Switch0

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

FastEthernet0/7

FastEthernet0/8

FastEthernet0/1

Port Status

Link Speed

Duplex

Access

VLAN

1

Tx Ring Limit

10

On

100 Mbps

10 Mbps

Auto

Half Duplex

Full Duplex

Auto

Equivalent IOS Commands

```

Switch(config-if)#exit
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/1
Switch(config-if)#shutdown
Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to administratively down.
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down.
Switch(config-if)#exit

```

*Untitled - Notepad

File Edit Format View Help

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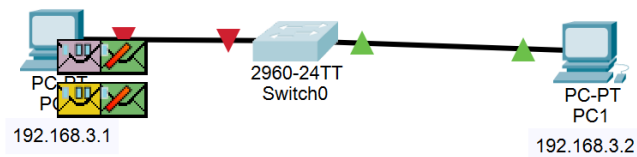
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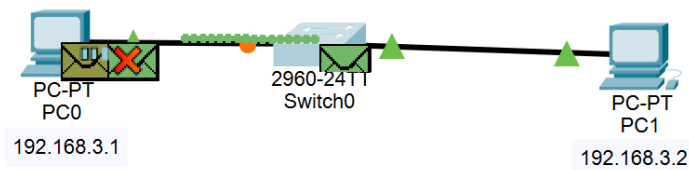
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Scenario 0

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What potential issues could arise if the sender's window size is too large compared to the network capacity, and how might this affect overall performance?

If the sender's window size is larger than network capacity, packets accumulate in routers and buffers overflow, causing packet loss.

Lost packets trigger retransmissions, which increases congestion, delay, and reduces overall throughput.

As a result, network performance degrades because bandwidth is wasted and both sender and receiver resources get overloaded.

Question 2: How does the fixed sliding window size of 4 in this simulation limit the accuracy of real-world protocol behavior, and what changes could be made to better reflect real network conditions?

A fixed window size of 4 cannot adapt to changing network conditions like bandwidth, delay, and error rate, so it may be too small for fast networks and too large for slow ones.

Because it has no congestion control, it can either waste bandwidth or cause packet loss and network congestion.

In real networks, protocols use dynamic window adjustment based on RTT, packet loss, and congestion feedback to keep transmission efficient.